

BanikAssignment6

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```
setwd("/Users/arpabanik/Library/CloudStorage/OneDrive-BentleyUniversity/R-Projects/MA 347/Assignment 6")
# Load the data
pharma.df <- read.csv("~/Library/CloudStorage/OneDrive-BentleyUniversity/R-Projects/MA 347/Assignment 6/Pharmaceutics.csv")

# Load required libraries
library(cluster)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
## filter, lag
```

```
## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union
```

```
library(readr)
```

1.

```
# set row names to pharma column
row.names(pharma.df) <- pharma.df[,1]
# Remove non-numeric variables
pharma_numeric <- pharma.df[, c(3:11)]
set.seed(347)
# Normalize input variables
pharma.norm.df <- sapply(pharma_numeric,scale)
# add row names
row.names(pharma.norm.df) <- row.names(pharma.df)
# compute normalized distance based on all 8 variables
d.norm <- dist(pharma.norm.df, method = "euclidean")
d.norm
```

```

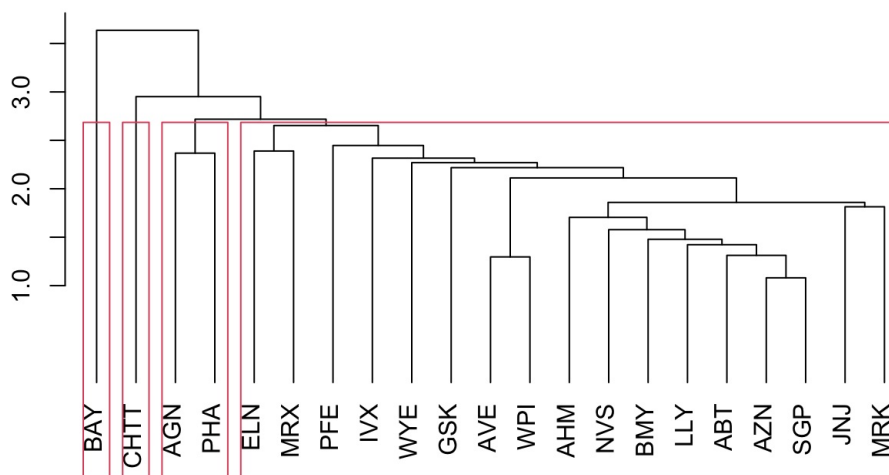
##          ABT          AGN          AHM          AZN          AVE          BAY          BMY          CHTT
## AGN 4.415575
## AHM 2.018793 3.945745
## AZN 1.669541 4.909566 2.364249
## AVE 2.111983 4.642699 2.487172 2.632282
## BAY 4.690231 4.853901 3.636353 5.065563 4.764654
## BMY 1.805543 5.419487 2.600986 1.572582 3.400602 5.273023
## CHTT 5.020726 5.612226 4.760341 5.719174 5.096246 4.969438 5.287400
## ELN 4.901141 6.695261 4.695844 4.974521 3.748778 4.608660 5.378092 4.675606
## LLY 1.422680 5.140253 3.238353 2.405951 2.910766 5.804419 2.189107 5.657801
## GSK 3.689906 6.747789 4.904614 2.957494 4.476690 7.546154 3.099023 7.080175
## IVX 2.624729 4.470028 2.316548 3.282195 2.386850 3.658011 3.279927 2.951511
## JNJ 2.333874 5.317942 3.593764 1.958326 3.640773 5.724303 2.511309 6.310233
## MRX 3.920297 5.479080 4.120549 4.269231 2.927258 4.848442 4.734766 4.786213
## MRK 2.680733 5.443918 3.361981 1.859280 3.472410 5.918477 2.432281 6.101541
## NVS 1.922731 5.468844 3.331743 3.056196 3.330879 5.331004 2.866126 6.063738
## PFE 3.887235 6.906828 5.268858 3.109413 4.495242 7.163993 3.666674 7.180257
## PHA 2.908982 2.367912 2.925627 3.715808 2.718441 3.955926 4.408645 5.000709
## SGP 1.312599 4.725384 1.704709 1.080519 2.464855 4.426418 1.478433 5.346513
## WPI 2.882610 5.007086 2.943946 3.414127 1.296549 5.055769 4.116074 5.540296
## WYE 3.038549 6.446458 4.185594 3.324966 4.254562 5.954379 2.269808 5.127981
##          ELN          LLY          GSK          IVX          JNJ          MRX          MRK          NVS
## AGN
## AHM
## AZN
## AVE
## BAY
## BMY
## CHTT
## ELN
## LLY 5.554227
## GSK 6.731204 3.631174
## IVX 3.115283 3.537378 5.276601
## JNJ 6.070533 2.722434 2.988672 4.354581
## MRX 2.389723 4.191466 6.187185 2.825394 5.306512
## MRK 5.921987 3.380695 2.218040 4.164267 1.814184 5.532520
## NVS 5.732322 1.577953 4.783039 3.899915 3.083678 4.478040 4.112418
## PFE 6.123133 3.783136 2.447177 5.356598 2.447341 5.518379 2.831329 4.536250
## PHA 5.007721 3.754900 5.773960 3.073579 4.112432 3.827019 4.448933 3.884035
## SGP 4.665611 2.205815 3.780283 2.763476 2.604437 3.907501 2.710607 2.542763
## WPI 3.756437 3.412378 5.437193 2.857109 4.591764 2.653341 4.569336 3.626404
## WYE 5.312455 2.747839 3.670720 3.719962 3.858028 4.709401 3.935039 3.525940
##          PFE          PHA          SGP          WPI
## AGN
## AHM
## AZN
## AVE
## BAY
## BMY
## CHTT
## ELN
## LLY
## GSK
## IVX
## JNJ
## MRX
## MRK
## NVS
## PFE
## PHA 5.587119
## SGP 3.955078 3.449579
## WPI 5.403128 3.172178 3.026610
## WYE 4.026095 5.286507 3.145472 4.922945

```

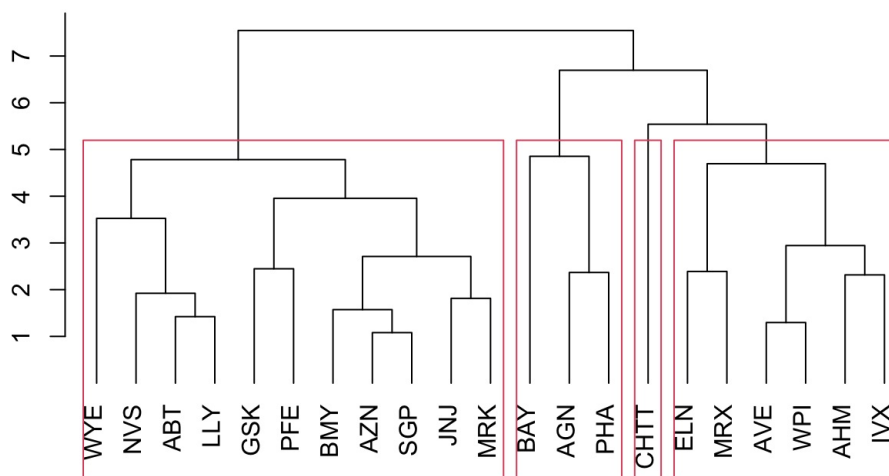
```

# Hierarchical Clustering
# Single Linkage
hc_single <- hclust(dist(pharma.norm.df), method = "single")
plot(hc_single, hang = -1, ann = FALSE)
rect.hclust(hc_single, 4)

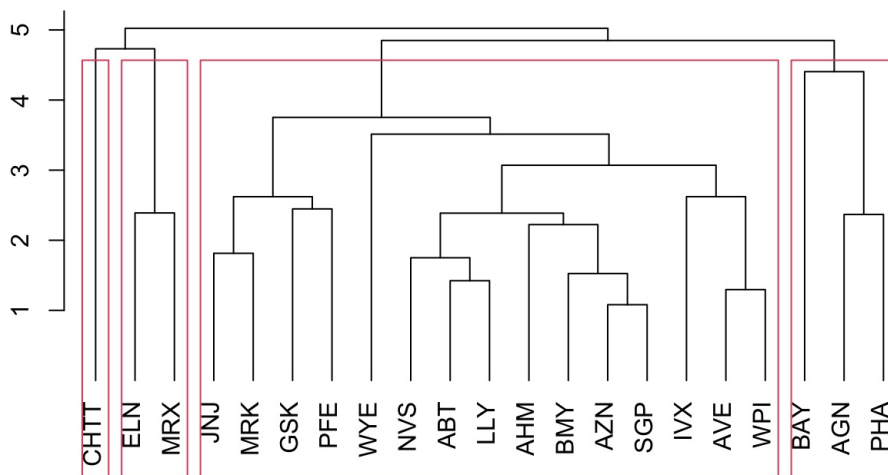
```



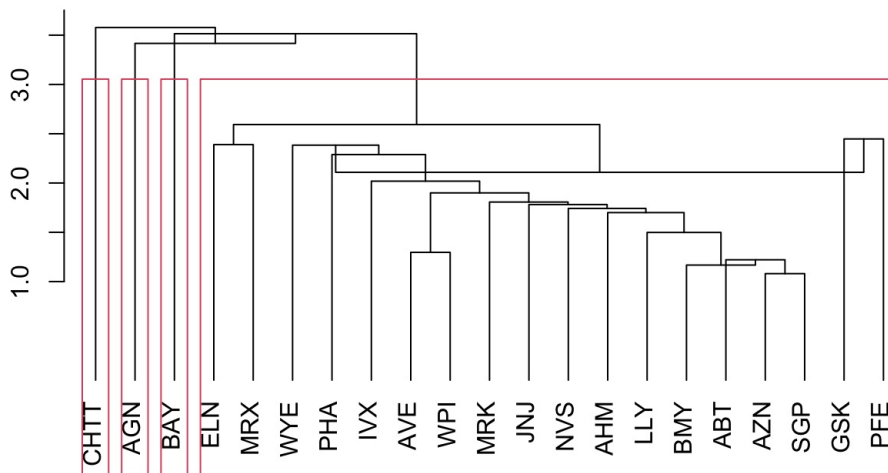
```
# Complete Linkage
hc_complete <- hclust(dist(pharma.norm.df), method = "complete")
plot(hc_complete, hang = -1, ann = FALSE)
rect.hclust(hc_complete, 4)
```



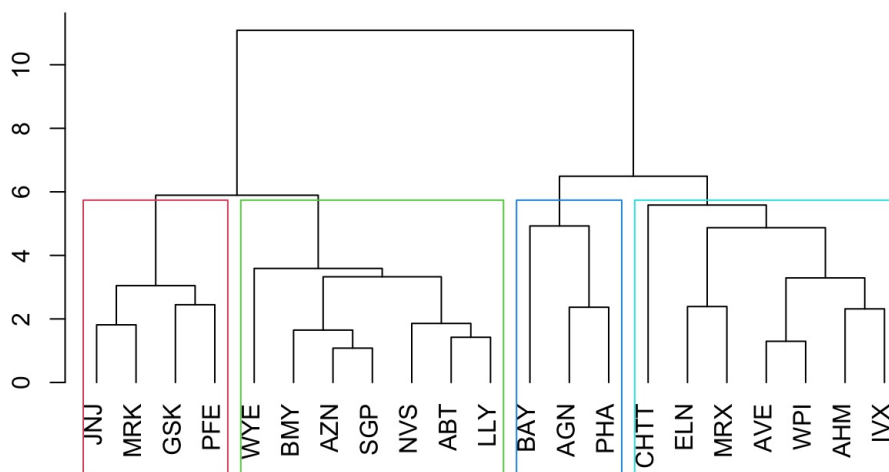
```
# Average Linkage
hc_average <- hclust(dist(pharma.norm.df), method = "average")
plot(hc_average, hang = -1, ann = FALSE)
rect.hclust(hc_average, 4)
```



```
# Centroids Linkage
hc_centroids <- hclust(dist(pharma.norm.df), method = "centroid")
plot(hc_centroids, hang = -1, ann = FALSE)
rect.hclust(hc_centroids, 4)
```



```
# Ward's Method
hc_ward <- hclust(dist(pharma.norm.df), method = "ward.D2")
plot(hc_ward, hang = -1, ann = FALSE)
rect.hclust(hc_ward, 4, border = 2:7)
```



I chose Ward's method with 4 clusters for this hierarchical clustering analysis of pharmaceutical firms' financial data because it minimizes within-cluster variance, ensuring that each cluster represents companies with similar financial characteristics, facilitating interpretation. Additionally, Ward's method tends to produce clusters of approximately equal sizes, promoting balanced assessments of different market segments. Its ability to create compact, spherical clusters enhances interpretability, crucial for equity analysts studying the pharmaceutical industry.

Let's summarize the memberships of the different clusters (which observations belong to which cluster):

```
memb1 <- cutree(hc_single, k = 4)
memb2 <- cutree(hc_complete, k = 4)
memb3 <- cutree(hc_average, k = 4)
memb4 <- cutree(hc_centroids, k = 4)
memb <- cutree(hc_ward, k = 4)
memb
```

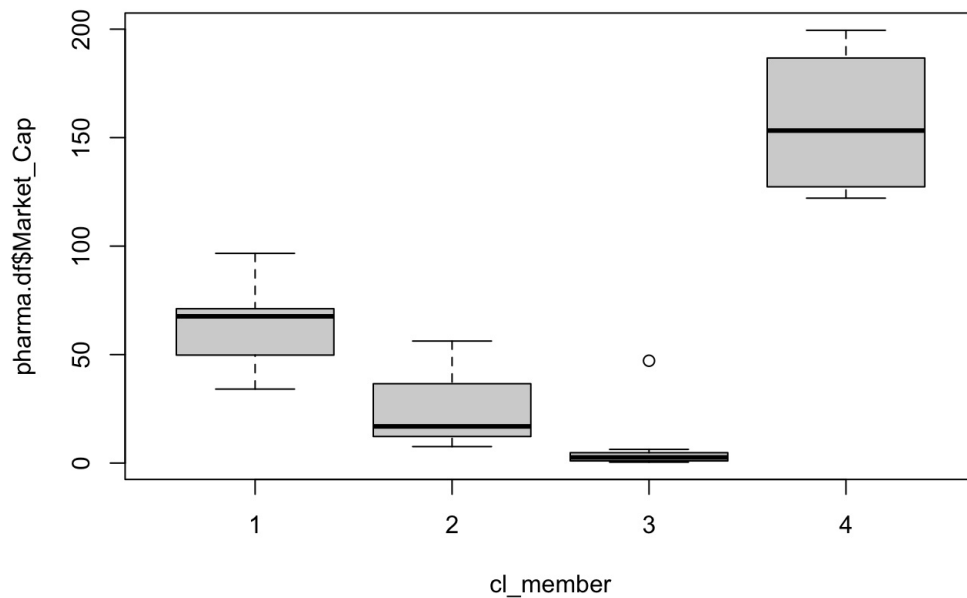
```
## ABT AGN AHM AZN AVE BAY BMY CHTT ELN LLY GSK IVX JNJ MRX MRK NVS
## 1 2 3 1 3 2 1 3 3 1 4 3 4 3 4 1
## PFE PHA SGP WPI WYE
## 4 2 1 3 1
```

2.

```
set.seed(347)
memb <- cutree(hc_ward, k = 4)
cl_member <- factor(memb)

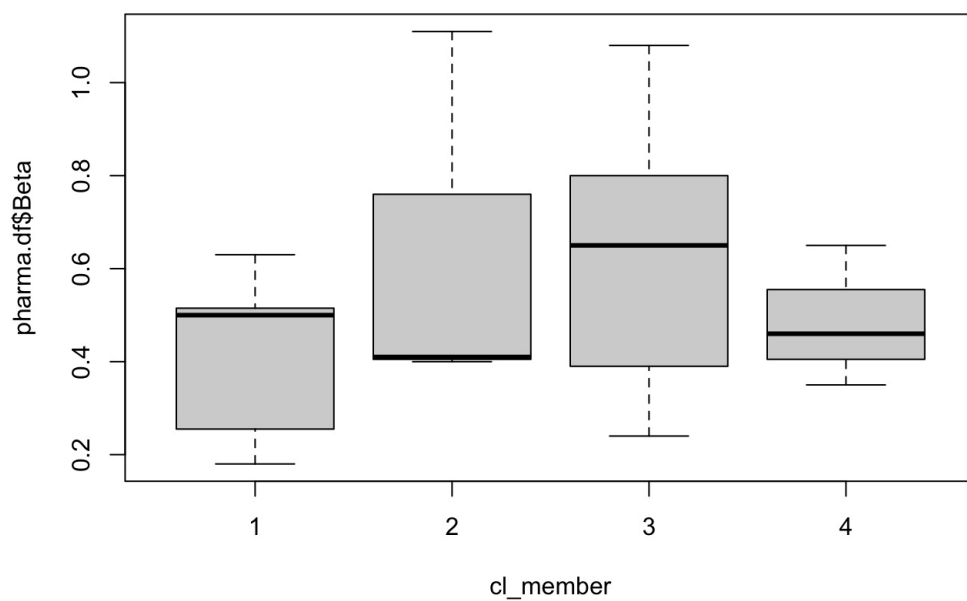
#Compare Market Cap (original scale) across four different clusters
boxplot(pharma.df$Market_Cap ~ cl_member, main="Market Cap Across Four Clusters")
```

Market Cap Across Four Clusters



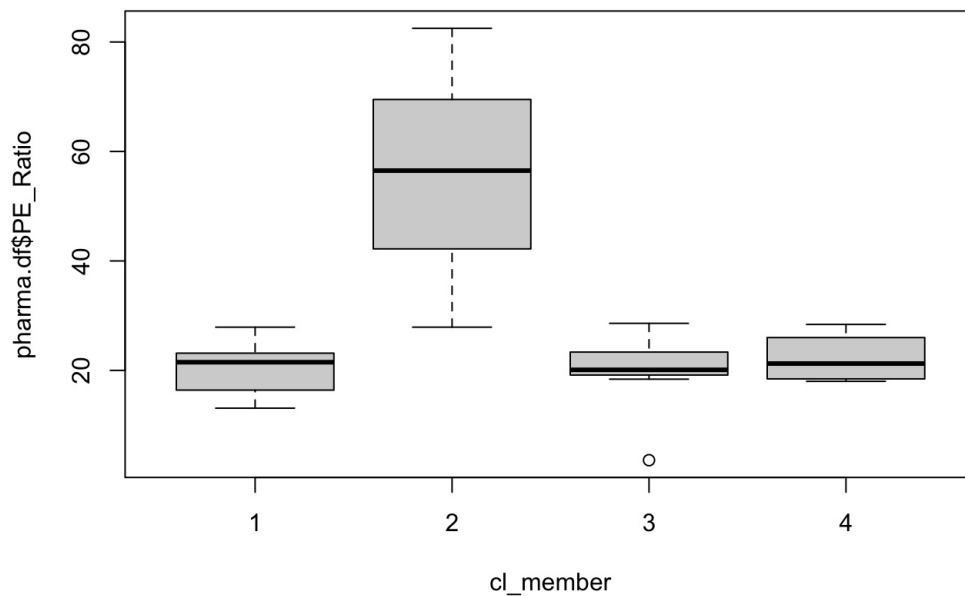
```
#Compare Beta (original scale) across four different clusters  
boxplot(pharma.df$Beta ~cl_member,main="Beta Across Four Clusters")
```

Beta Across Four Clusters



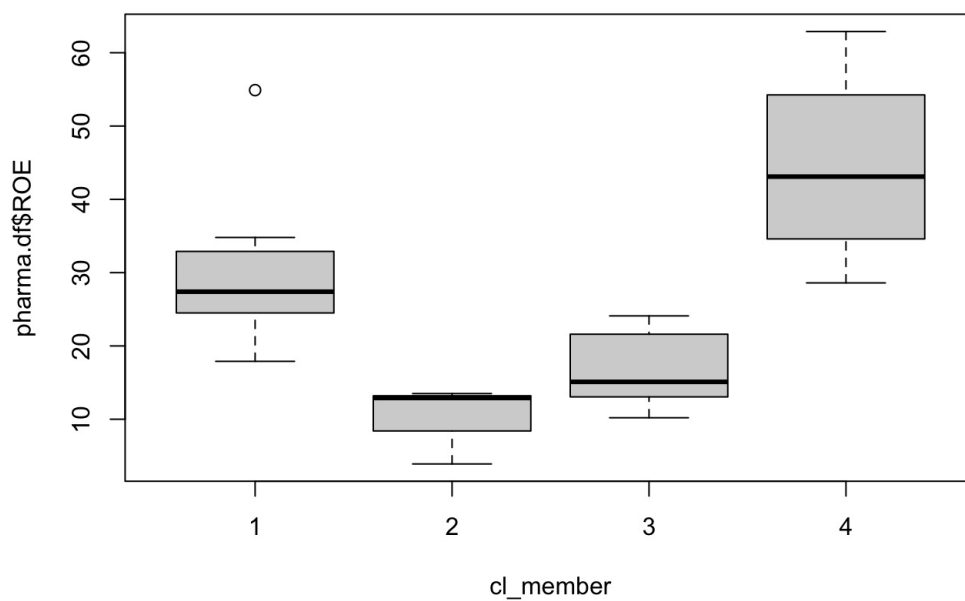
```
#Compare PE Ratio (original scale) across four different clusters  
boxplot(pharma.df$PE_Ratio ~cl_member,main="PE Ratio Across Four Clusters")
```

PE Ratio Across Four Clusters



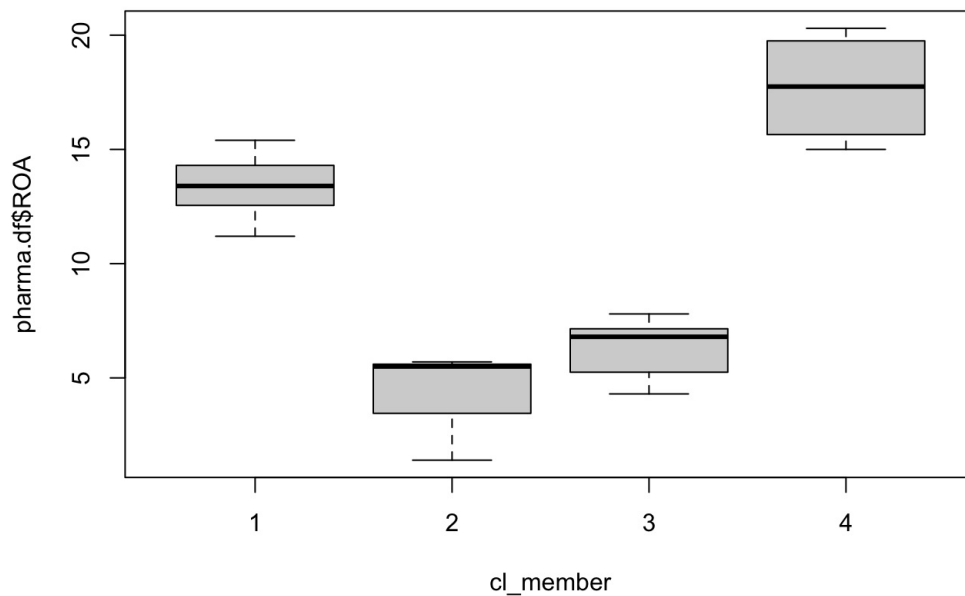
```
#Compare ROE (original scale) across four different clusters  
boxplot(pharma.df$ROE ~cl_member,main="ROE Across Four Clusters")
```

ROE Across Four Clusters



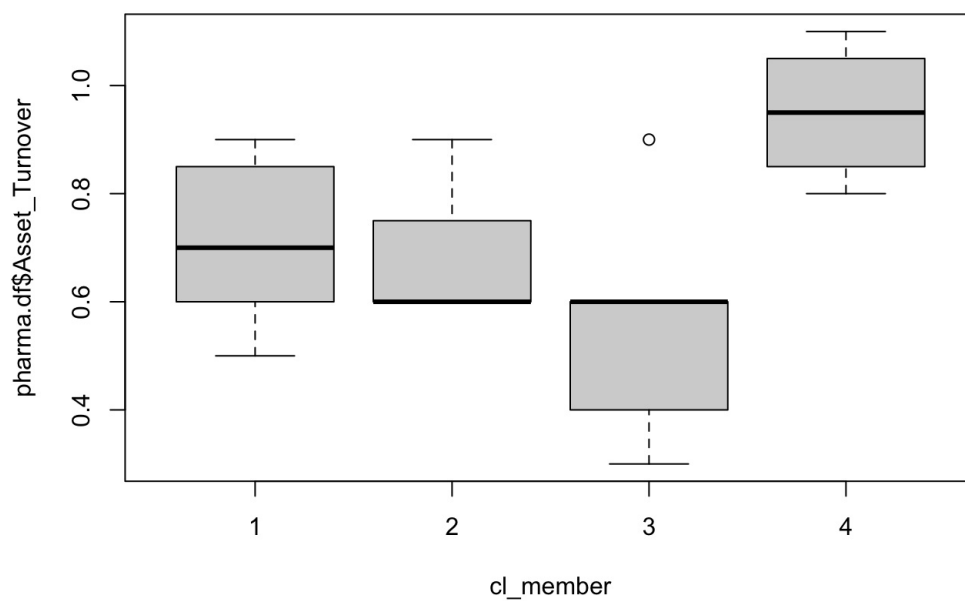
```
#Compare ROA (original scale) across four different clusters  
boxplot(pharma.df$ROA ~cl_member,main="ROA Across Four Clusters")
```

ROA Across Four Clusters



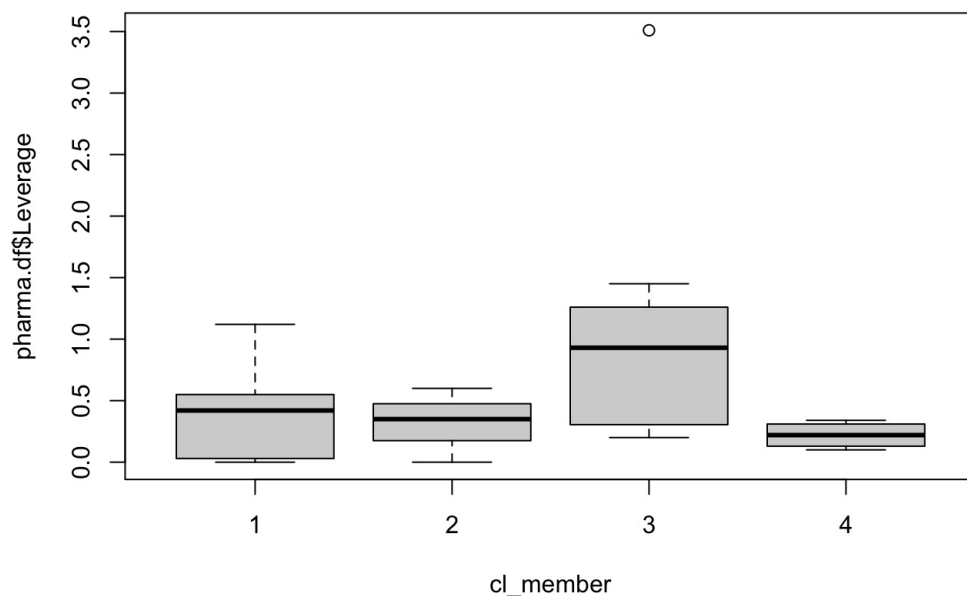
```
#Compare Asset Turnover (original scale) across four different clusters  
boxplot(pharma.df$Asset_Turnover ~cl_member,main="Asset Turnover Across Four Clusters")
```

Asset Turnover Across Four Clusters



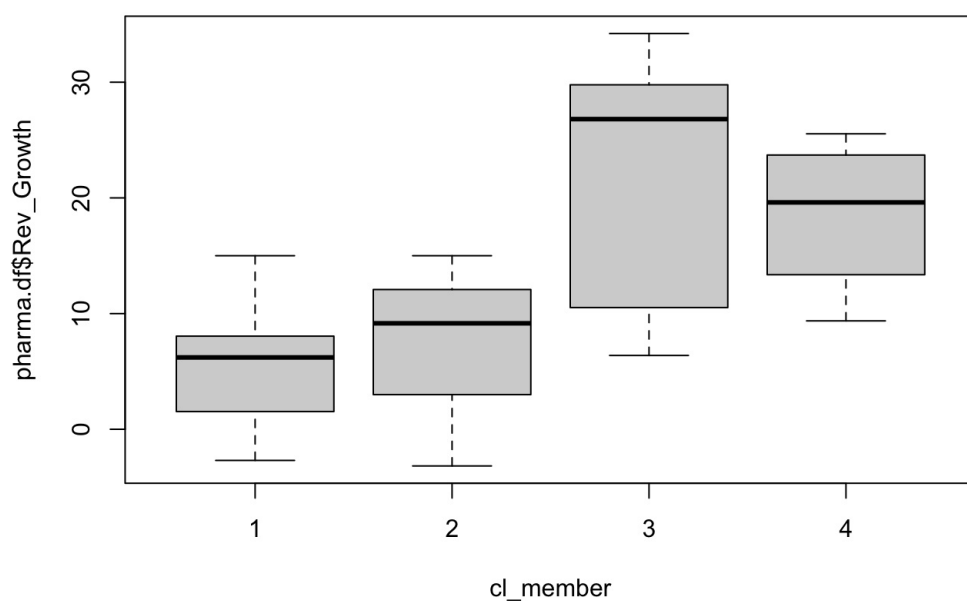
```
#Compare Leverage (original scale) across four different clusters  
boxplot(pharma.df$Leverage ~cl_member,main="Leverage Across Four Clusters")
```


Leverage Across Four Clusters



```
#Compare Rev_Growth (original scale) across four different clusters
boxplot(pharma.df$Rev_Growth ~cl_member,main="Revenue Growth Across Four Clusters")
```

Revenue Growth Across Four Clusters



This cluster exhibits the highest values for the net profit margin compared to the other clusters. Additionally, it shows relatively high values for return on assets (ROA), while leverage and revenue growth are comparatively lower. These characteristics suggest a group of firms with strong profitability metrics and efficient asset utilization, albeit with conservative financial leverage and moderate revenue growth.

Cluster 2 demonstrates the highest values for beta and the price/earnings ratio, although both exhibit high variance within the cluster. Conversely, net profit margin, return on equity (ROE), and ROA are relatively low with low variance. This indicates a cluster with firms characterized by higher market risk and valuation multiples but with varying levels of profitability.

Firms in Cluster 3 have high values for leverage and revenue growth, with notable variance observed in these variables. However, market capitalization and asset turnover values are comparatively low. This cluster represents companies with higher financial leverage and growth potential, although they may face challenges in efficiently utilizing assets to generate revenue and market value.

Cluster 4 features the highest values for market capitalization, ROE, ROA, and asset turnover. Conversely, beta, price/earnings ratio, and leverage variables exhibit lower values, with consistent low variance. This cluster suggests firms with strong market presence, profitability, and asset efficiency, coupled with lower market risk and financial leverage.

We can also add numeric summary of each cluster:

```
clusmeans<-aggregate(.~memb,data=pharma_numeric,FUN=mean)
clusmeans
```

```
##      memb Market_Cap      Beta PE_Ratio      ROE      ROA Asset_Turnover
## 1      1      62.882857 0.4071429 20.22857 30.71429 13.385714      0.7142857
## 2      2      26.906667 0.6400000 55.63333 10.10000 4.200000      0.7000000
## 3      3       8.815714 0.6214286 19.61429 16.95714 6.242857      0.5428571
## 4      4     157.017500 0.4800000 22.22500 44.42500 17.700000      0.9500000
##      Leverage Rev_Growth Net_Profit_Margin
## 1 0.3857143      5.382857      20.514286
## 2 0.3166667      6.996667      5.133333
## 3 1.1100000     21.141429      13.185714
## 4 0.2200000     18.532500      19.575000
```

```
clusvars<-aggregate(.~memb,data=pharma_numeric,FUN=var)
clusvars
```

```
##      memb Market_Cap      Beta PE_Ratio      ROE      ROA Asset_Turnover
## 1      1      415.2496 0.03132381 29.16238 143.56143 2.384762      0.02476190
## 2      2       667.0489 0.16570000 745.85333 28.92000 5.890000      0.03000000
## 3      3      289.8903 0.09038095 63.49143 30.11619 1.736190      0.04285714
## 4      4     1301.6501 0.01553333 22.97583 202.58917 6.086667      0.01666667
##      Leverage Rev_Growth Net_Profit_Margin
## 1 0.16646190      34.25316      11.841429
## 2 0.09083333      86.04723      5.623333
## 3 1.33936667     136.71941      18.441429
## 4 0.01200000      48.53123      22.249167
```

```
clusSD<-aggregate(.~memb,data=pharma_numeric,FUN=sd)
clusSD
```

```
##      memb Market_Cap      Beta PE_Ratio      ROE      ROA Asset_Turnover
## 1      1      20.37767 0.1769853 5.400220 11.981712 1.544267      0.1573592
## 2      2      25.82729 0.4070626 27.310316 5.377732 2.426932      0.1732051
## 3      3      17.02617 0.3006343 7.968151 5.487822 1.317646      0.2070197
## 4      4      36.07839 0.1246328 4.793311 14.233382 2.467117      0.1290994
##      Leverage Rev_Growth Net_Profit_Margin
## 1 0.4079974      5.852620      3.441138
## 2 0.3013857      9.276165      2.371357
## 3 1.1573101     11.692708      4.294348
## 4 0.1095445      6.966436      4.716902
```

3.

```
set.seed(347)
# Analyze categorical variables across clusters
# Median Recommendation
Market_Recommendation<-table(memb, pharma.df[,12])
Market_Recommendation
```

```
##
## memb Hold Moderate Buy Moderate Sell Strong Buy
##      1      4      1      2      0
##      2      2      1      0      0
##      3      1      3      2      1
##      4      2      2      0      0
```

```
# Location
Location<-table(memb, pharma.df[,13])
Location
```

```
##
## memb CANADA FRANCE GERMANY IRELAND SWITZERLAND UK US
##      1      0      0      0      0      1 1 5
##      2      1      0      1      0      0 0 1
##      3      0      1      0      1      0 1 4
##      4      0      0      0      0      0 1 3
```

```
# Exchange
Exchange<-table(memb, pharma.df[,14])
Location
```

```
##
## memb CANADA FRANCE GERMANY IRELAND SWITZERLAND UK US
## 1 0 0 0 0 1 1 5
## 2 1 0 1 0 0 0 1
## 3 0 1 0 1 0 1 4
## 4 0 0 0 0 0 0 3
```

Median Recommendation: Cluster 1 has a mix of recommendations, with a slightly higher tendency towards “Moderate Buy” and “Moderate Sell”. Cluster 2 mostly consists of “Moderate Buy” recommendations. Cluster 3 shows a diverse range of recommendations, including “Hold”, “Moderate Buy”, “Moderate Sell”, and “Strong Buy”. Cluster 4 predominantly consists of “Moderate Buy” and “Hold” recommendations. Location: Cluster 1 has firms primarily from the US, with one from Switzerland and one from the UK. Cluster 2 has a more diverse geographical spread with firms from Canada, Germany, and the US. Cluster 3 also shows diversity, including firms from France, Ireland, Switzerland, the UK, and the US. Cluster 4 mainly consists of firms from the US. Exchange: Cluster 1 mainly lists its firms on the NYSE, with one firm from Switzerland. Cluster 2 has firms listed on various exchanges including NYSE, TSX, and FWB. Cluster 3 primarily lists its firms on exchanges in France, Germany, Ireland, the UK, and the US. Cluster 4 mostly lists its firms on the NYSE, with one firm from the NASDAQ.

In summary, while the patterns may not be perfectly distinct, there are observable trends in the clusters, particularly regarding the geographical location of firms especially in the US and the types of recommendations they receive.

4. Diversified Performers: This cluster consists of firms with a mix of recommendations and a diverse geographical spread, primarily listed on the NYSE. They exhibit moderate financial performance across various metrics. Stable Performers: Firms in this cluster receive mostly “Moderate Buy” recommendations and are listed on various exchanges. They demonstrate stable financial performance and moderate market capitalization. Global Leaders: This cluster includes firms with diverse recommendations and a global presence, listed on exchanges across different countries. They exhibit strong financial performance and substantial market capitalization. US Dominance: Firms in this cluster are primarily based in the US and listed on the NYSE, with a mix of recommendations leaning towards “Moderate Buy”. They demonstrate solid financial performance and significant market capitalization within the US market.

5.

```
# Perform k-means clustering
set.seed(347) # for reproducibility
kmeans_result <- kmeans(pharma_numeric, centers = 4)

# Display cluster membership
cluster_membership <- kmeans_result$cluster
cluster_membership
```

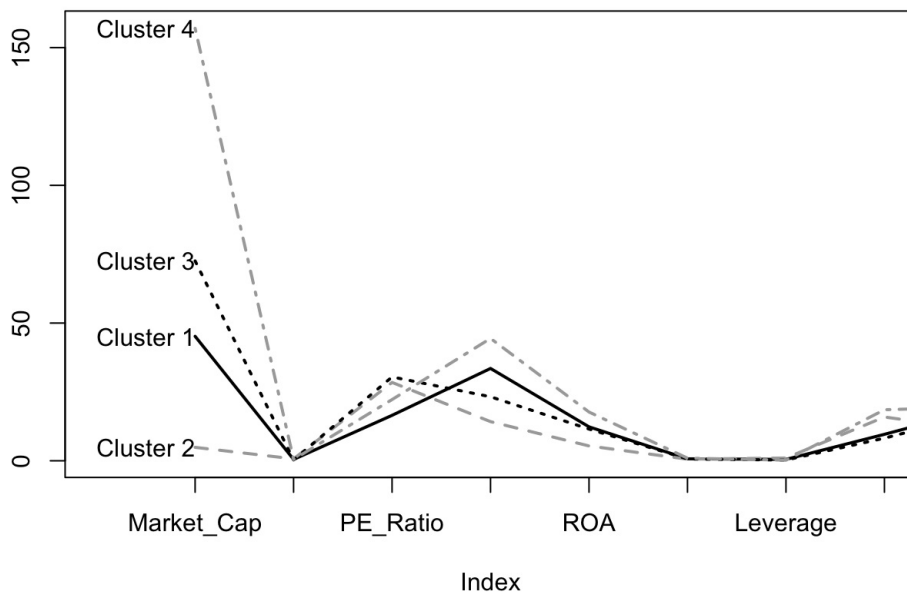
```
## ABT AGN AHM AZN AVE BAY BMY CHTT ELN LLY GSK IVX JNJ MRX MRK NVS
## 3 2 2 3 1 2 1 2 2 3 4 2 4 2 4 3
## PFE PHA SGP WPI WYE
## 4 3 1 2 1
```

```
# plot an empty scatter plot
plot(c(0), xaxt = 'n', ylab = "", type = "l",
      ylim = c(min(kmeans_result$centers), max(kmeans_result$centers)), xlim = c(0, 8))

# label x-axes
axis(1, at = c(1:9), labels = names(pharma_numeric))

# plot centroids
for (i in c(1:4))
  lines(kmeans_result$centers[i,], lty = i, lwd = 2, col = ifelse(i %in% c(1, 3, 5),
                                                                    "black", "dark grey"))

# name clusters
text(x = 0.5, y = kmeans_result$centers[, 1], labels = paste("Cluster", c(1:4)))
```



While there are some differences in the assignments of individual firms to clusters between k-means and hierarchical clustering, the overall picture remains somewhat consistent. Both methods identify clusters with similar characteristics and tendencies, such as grouping together certain firms based on their financial metrics. However, there are variations in the specific firms assigned to each cluster, which is expected due to differences in the algorithms and approaches of the two clustering methods.